

1. INTRODUCTION CHAPTER 1

Should be like any other project

2. LITERATURE REVIEW

This is what I expect to be in the literature Review

- i. Analyse existing software or solutions to discuss features, strengths & weaknesses for each system.
- ii. Discuss relevant technologies, tools, frameworks, programming languages, databases, cloud services, etc.
- iii. As a research gap, explain what existing systems fail to address and how your artefact will improve on them.

3. RESEARCH METHODOLOGY

Briefly explain Design Science Research including problem identification, define objectives, design, development, demonstration and evaluation. You can include a diagram of the DSR process.

- i. How you did the requirements gathering- to include methods that you used like interviews, observation, questionnaires and document analysis
- ii. How you did the requirements analysis including functional requirements, non-functional requirements, user requirements, system requirements

System Design section should include

- i. system architecture and database design (ER Diagram) etc
- ii. UML Diagrams (use case diagram, class diagram, activity diagram & sequence diagram)
- iii. Interface Design including wireframes or screen mock-ups.
- iv. Algorithm or Process Design including flowcharts or pseudocode.
- v. Technology Stack - Explaining why the selected technologies were chosen.

4. CHAPTER FOUR: SYSTEM DEVELOPMENT, DEMONSTRATION AND EVALUATION

- i. Under System Development, explain implementation (programming language, database, framework, development environment)
- ii. Under System Demonstration, describe how users perform key tasks, including screenshots.
- iii. Under testing, describe whether you did (unit testing, integration testing,
- iv. system testing, User Acceptance Testing (UAT)) - Present results in tables.
- v. Evaluation is a key DSR component.
- vi. Evaluate whether the artifact solves the identified problem. You can do that using any of these
 - a. Expert review

- b. User testing
- c. Usability testing
- d. Performance testing
- e. Questionnaire-based user satisfaction
- f. Comparison with existing systems
- vii. After testing you discuss effectiveness, efficiency, usability, reliability, security (if applicable)

5. CHAPTER FIVE: SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

Summarise the study.

- i. Explain how the objectives were achieved.
- ii. Describe the practical contributions of the developed artefact.
- iii. Suggest improvements.
- iv. Discuss possible extensions as future work.

REFERENCES

Use an approved citation style (e.g., APA 7th edition).

APPENDICES

May include things like an interview guide, questionnaire, source code (selected sections), user manual, test cases and additional screenshots

Mapping the Report to the Design Science Research Process

DSR Phase	Report Chapter
Problem Identification	Chapter 1
Define Objectives	Chapter 1
Knowledge Base/Literature Review	Chapter 2
Design	Chapter 3
Development	Chapter 4
Demonstration	Chapter 4
Evaluation	Chapter 4